SOMIN

ITMO UNIVERSITY

LAB 1

LinkedIn Post Engagement Prediction

Navella Sevumyan

> **Objective:** Predict whether a LinkedIn post will perform below, near, or above its author's usual engagement.

Task & Dataset

89,018
Labeled Posts

71,395
Training Posts

17,623
Test Posts

Dataset & Evaluation:

- Author-disjoint split
- Three engagement classes
- Macro-F1 as the primary metric
- QWK as a secondary metric

Our Goal:

Predict whether a LinkedIn post performs below, near, or above its author's usual engagement.

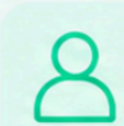
FEATURE ENGINEERING · OVERVIEW

Feature Engineering

01 — Historical Engagement

Rolling averages, standard deviations, and engagement trends.

02 — Author-Relative Features



Post length, posting patterns, and lexical novelty relative to previous posts.

03 — Text Representation

Word TF-IDF and Character TF-IDF (3–5 grams).

Takeaway: Combining post-level features with author history provides a richer representation than metadata alone.

MODELS & EXPERIMENTS · OVERVIEW

Models & Experiments

Model Selection

LATE FUSION

One model per source, then combine

MODEL SELECTION
V3 improved validation Macro-F1 from 0.6486 to 0.6501 (+0.0015), below our predefined threshold of 0.0025. Therefore, we retained V2.

EXPERIMENTAL DECISIONS · MODEL DEVELOPMENT

Why I Chose My Final Approach

01

Starting Point

Started with CatBoost on structured post and author features as a baseline.

→

02

Adding Text

Added Word TF-IDF and later Character TF-IDF to capture lexical patterns.

→

03

Author Context

Added historical and author-relative features to model typical behaviour.

→

04

Model Selection

Tested V3, but the improvement was too small, so V2 was retained.

WHY THESE CHANGES?

- **Structured features:** Capture metadata and engagement patterns.
- **Text features:** Add lexical information missing from tabular data.
- **Author context:** Account for differences in authors' typical engagement.

KEY DECISION

+0.0015 Macro-F1

V3's improvement was below our 0.0025 threshold, so V2 was retained.

SOURCE Internal experiments; model versions V2 and V3; evaluation on the validation split, Macro-F1.

RESULTS · LAB 1

Fusion improved the final score over the baseline.

MACRO-F1

SUMMARY METRICS

Model	Macro-F1	QWK
Baseline	0.5410	0.4576
V1	0.6248	0.5474
Final	**0.6319**	**0.5612**

+16.8%

Relative Macro-F1 improvement over the baseline.

SOURCE Own experiments on the Kairos dataset.

CONCLUSIONS · KEY TAKEAWAYS

Author-aware features and model combination improve engagement prediction.

- **Author history** provides valuable engagement context.
- **Combining tabular and textual models** improves prediction quality.
- **Word and character TF-IDF** capture complementary signals.
- **Increasing ensemble complexity** did not produce a substantial validation improvement.

FINAL RESULT

0.6319 Macro-F1

0.5612 QWK

Future work: semantic retrieval, image features, and social graph information.